

Name:

Instructions: Homework is graded primarily on correctness and completion.

- Write your solutions on a separate sheet of paper.
- Include all key steps and explanations that justify your answers, unless it says otherwise.
- If you print this homework set, you may write directly on the page where appropriate.
- Attach all supporting files (e.g. .py or .ipynb files) when needed.

Homework Set 4

1.

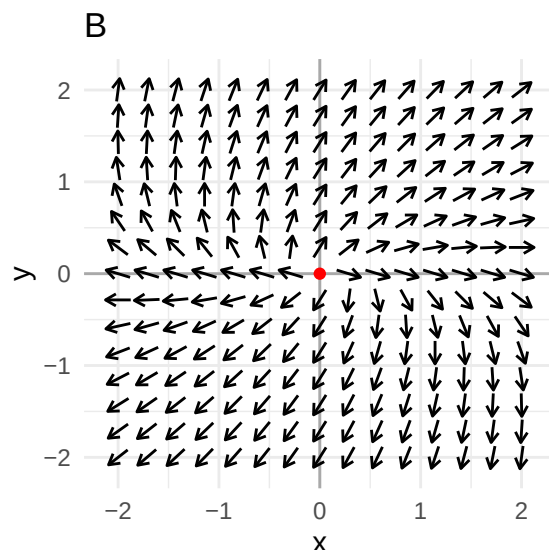
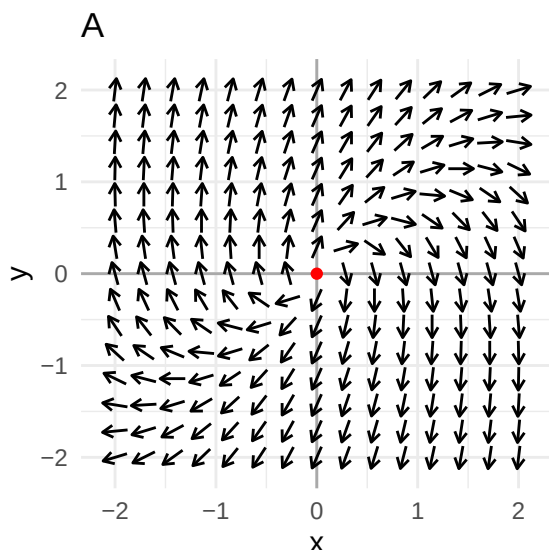
Let x and y be the dependent variables with t as the independent variable. For each system use the given information to:

- determine the general and specific solution in vector form,
- identify the correct direction field, shown below, by matching the eigenvectors,
- draw the specific solution on the direction field, and
- use a graphing technology to plot the specific solution with respect to the t axis.

Then, describe the similarities and differences between how solution behavior appears on the phase plane and how it appears when graphed as a function of time.

a) $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}, \quad \vec{y}(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \quad \lambda_1 = 5, \quad \vec{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

b) $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}, \quad \vec{y}(0) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}; \quad \lambda_1 = 3 + 2i, \quad \vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + i \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + i \begin{bmatrix} 1 \\ 0 \end{bmatrix}$



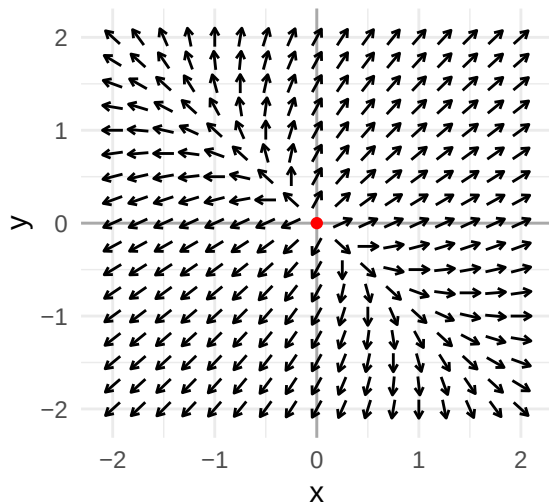
2.

Consider the following systems of ODEs with their corresponding direction fields:

System A

$$x' = 2x + y$$

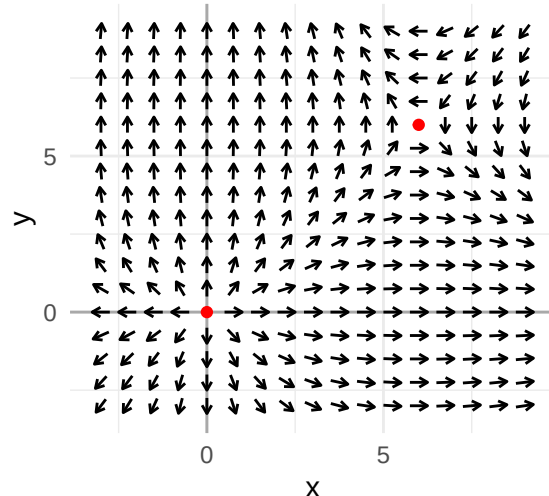
$$y' = x + 2y$$



System B

$$x' = 2x - \frac{1}{3}xy$$

$$y' = 3y - \frac{1}{2}xy$$



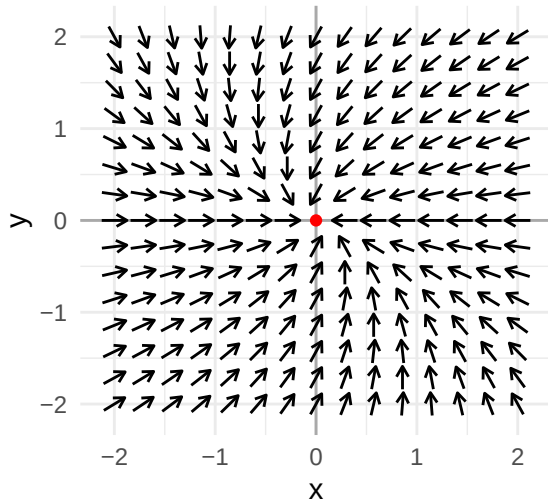
- Determine the Jacobian matrices $J_A(x, y)$ and $J_B(x, y)$ for systems A and B respectively. Compare the matrices and identify which system is linear and non-linear. Then, explain the key indicator of the system's linearity based on the matrices.
- For each system, determine the equilibria and evaluate their Jacobian matrix at those points. Then, explain the purpose of the Jacobian matrix in the process of linearization.
- For each system, determine the eigenvalues of each Jacobian matrix. Then, explain what the eigenvalues describe regarding the behavior of solutions around each equilibrium.
- For the linear system only:
 - Determine the "straight-line" solutions using the isocline $\frac{y'}{x'} = \frac{y}{x}$ and compare these solutions to the eigenvalues for the linear system computed in Part (d). Describe your observations.
 - Draw these "straight-line" solutions on the system's corresponding direction field. Describe the behavior of solutions around the equilibrium.
 - Are the properties of the eigenvalues consistent with your observations at the equilibrium?
- For the non-linear system only:
 - Explain why the linearization process is necessary to determine the behavior of solutions around an equilibrium.
 - Draw at least four specific solution curves on the system's corresponding direction field. Describe the behavior of solutions around each equilibrium.
 - Are the properties of the eigenvalues consistent with your observations at each equilibrium?

3.

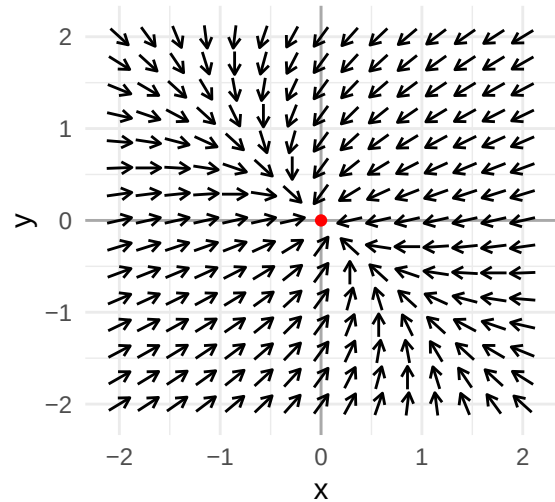
Let x and y be the dependent variables with t as the independent variable. For each of the following homogeneous linear system of 1st-order ODEs:

- (i) determine the general solution,
- (ii) on its corresponding direction field, identify the “straight-line” solutions if any,
- (iii) on the same direction field, draw at least four specific solution curves, and
- (iv) identify the equilibrium class (including the relevant subclass) and its stability.

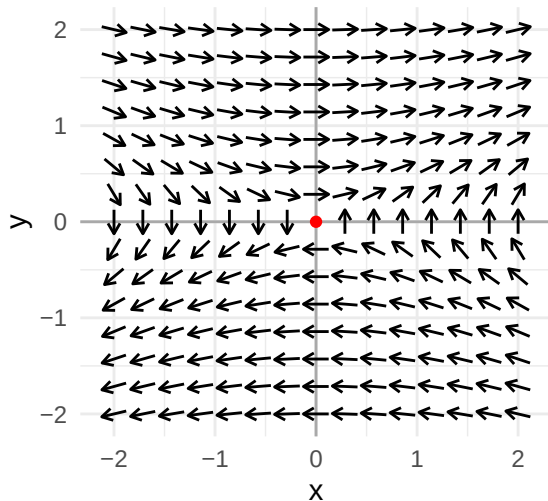
a)
$$\begin{aligned} x' &= -2x - 1y \\ y' &= -2y \end{aligned}$$



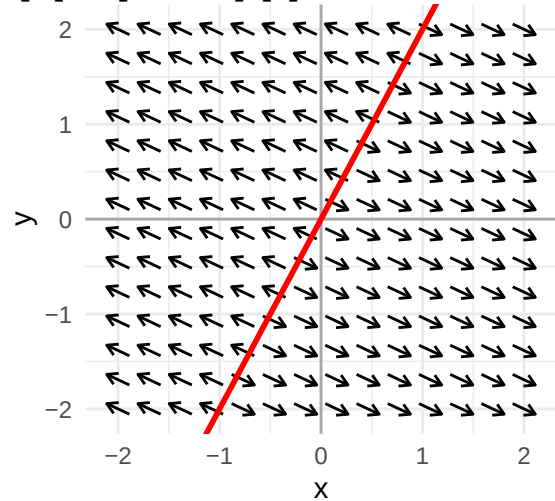
b)
$$\begin{aligned} x' &= -4x - 2y \\ y' &= -x - 3y \end{aligned}$$



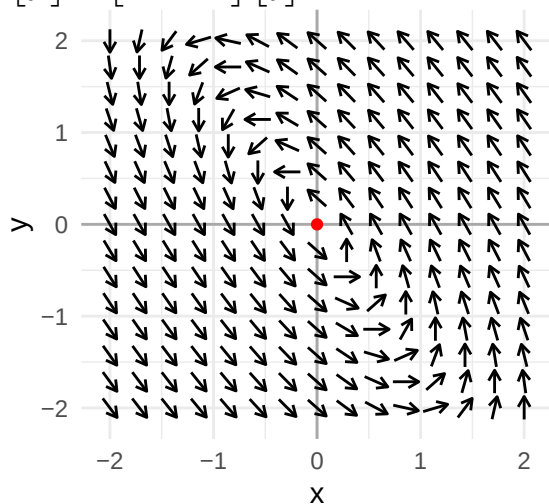
c)
$$\begin{aligned} x' &= 4y \\ y' &= x \end{aligned}$$



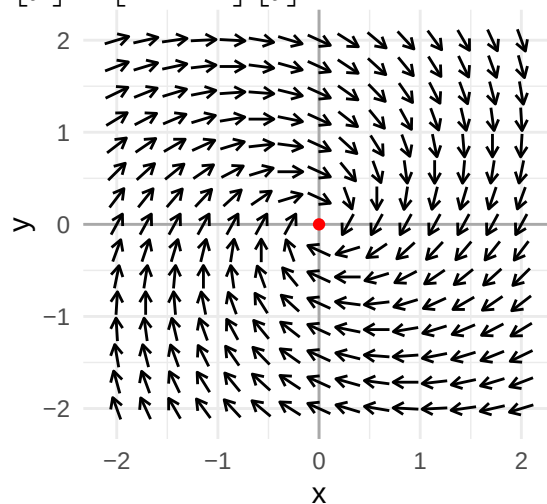
d)
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



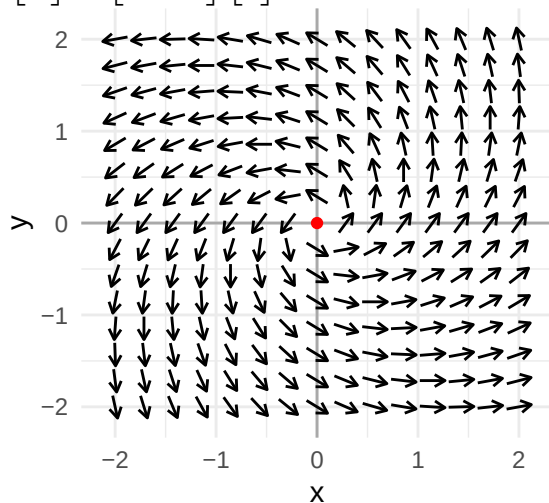
$$e) \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



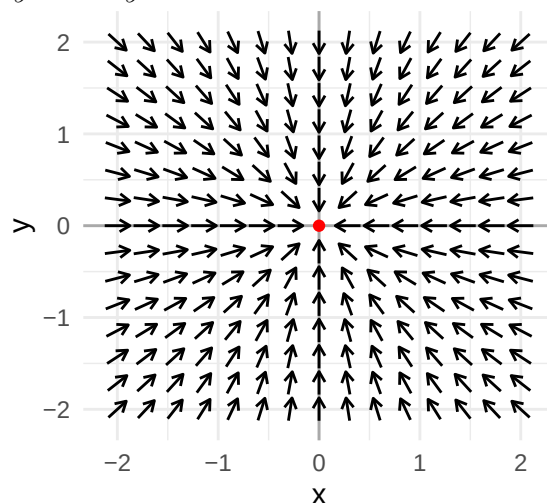
$$f) \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



$$g) \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



$$h) \begin{aligned} x' &= -2x \\ y' &= -2y \end{aligned}$$



4. Challenge Problem

Consider the following homogeneous linear system of 1st-order ODEs.

$$\begin{aligned}x' &= -\alpha x + \beta y \\ y' &= \beta x - \alpha y\end{aligned}$$

where $\alpha > 0$ and $\beta > 0$.

- a) Determine the equation for the x -nullcline of the system of ODEs with terms of α and β .
- b) Determine the equation for the y -nullcline of the system of ODEs with terms of α and β .
- c) Determine the "straight-line" solutions (eigenvalues and eigenvectors) with terms of α and β in two ways:

- i. Using the isocline $\frac{y'}{x'} = \frac{y}{x}$.

- ii. Using the matrix-vector form $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -\alpha & \beta \\ \beta & -\alpha \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$.

Then, compare the results obtained by each method. Discuss any similarities or differences.

- d) Create complete and well represented phase portraits by hand with at least four specific solution curves for cases when $\alpha = \beta$, $\alpha < \beta$, and $\alpha > \beta$.
 - e) Describe the behavior of solutions around the equilibrium for cases when $\alpha = \beta$, $\alpha < \beta$, and $\alpha > \beta$.